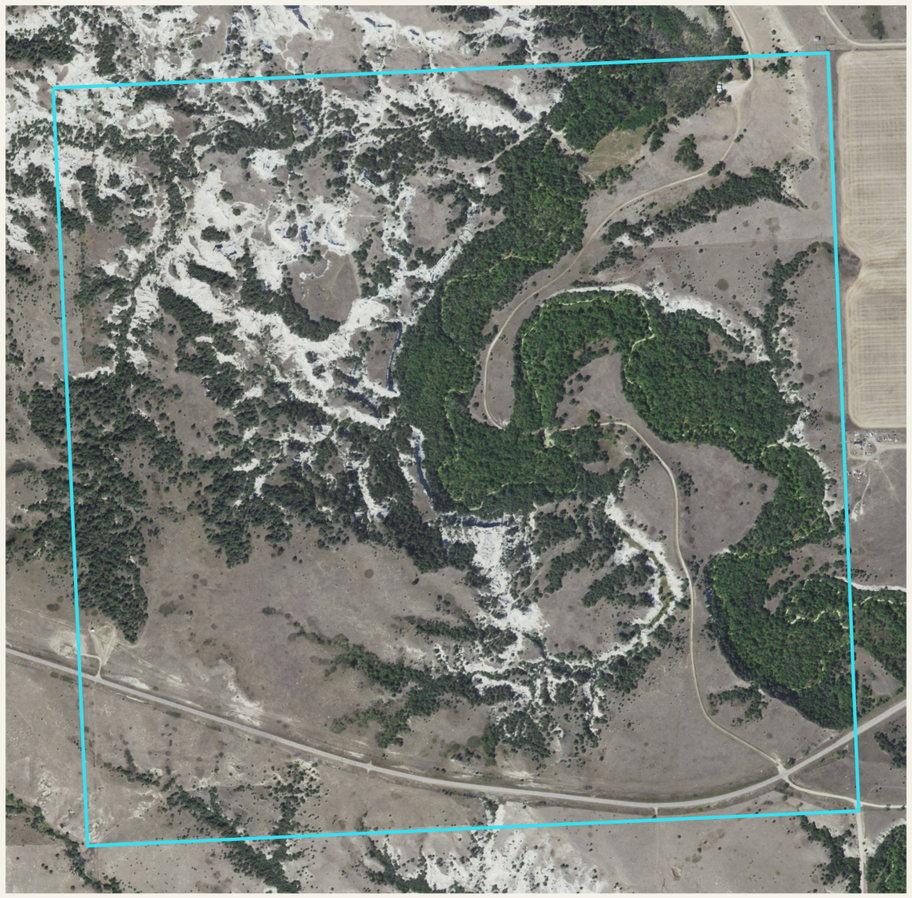
ATE MACOKE

Understanding Our Land

Community Land Assessment Mapbook

ATE_MACOKE
A VIEW OF THE LAND



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

Environmental Data Cube for Oglala Lakota
2015–2023 | Community Review Draft

Reading the Land Together

This mapbook brings together satellite observations developed for the Ate Macoke land assessment.

The maps offer different views of vegetation, land cover, and changes through time. They are meant to support conversation and closer observation.

Knowledge of the land is essential. Weather, seasonal conditions, plant communities, and land use can all affect what satellite images show.

Each map includes an explanation and a question for discussion. The mapbook can be used without internet access.

MAP 01

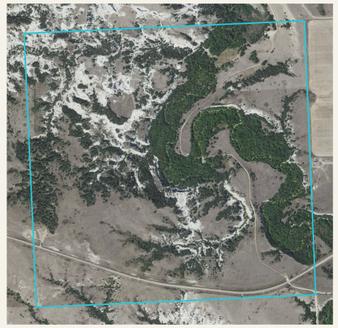
How Is My Land Doing?

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

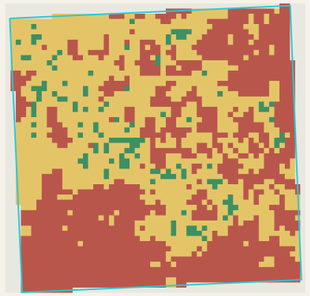
ATE_MACOKE | HOW IS MY LAND DOING?

Aerial photography, vegetation change, and land-cover information

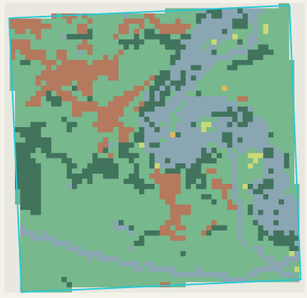
A / WHAT THE LAND LOOKS LIKE



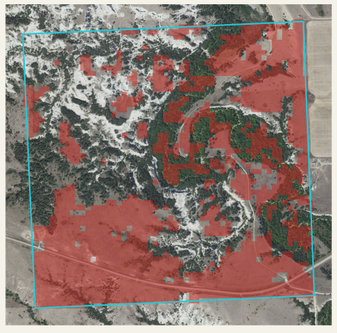
B / HOW VEGETATION CHANGED



C / WHAT KIND OF LAND IS IT?



D / WHERE SHOULD WE LOOK CLOSER?



VEGETATION CHANGE
Less green · Little change · Greener

LAND-COVER TYPES
Grassland / Pasture · Cultivated Crops · Forest / Shrubland
Hay / Alfalfa · Barren · Other Land

Red areas show reduced vegetation greenness from 2015 to 2023, not confirmed land degradation. Aerial imagery date may differ from the NDVI comparison years.

READING THE MAP

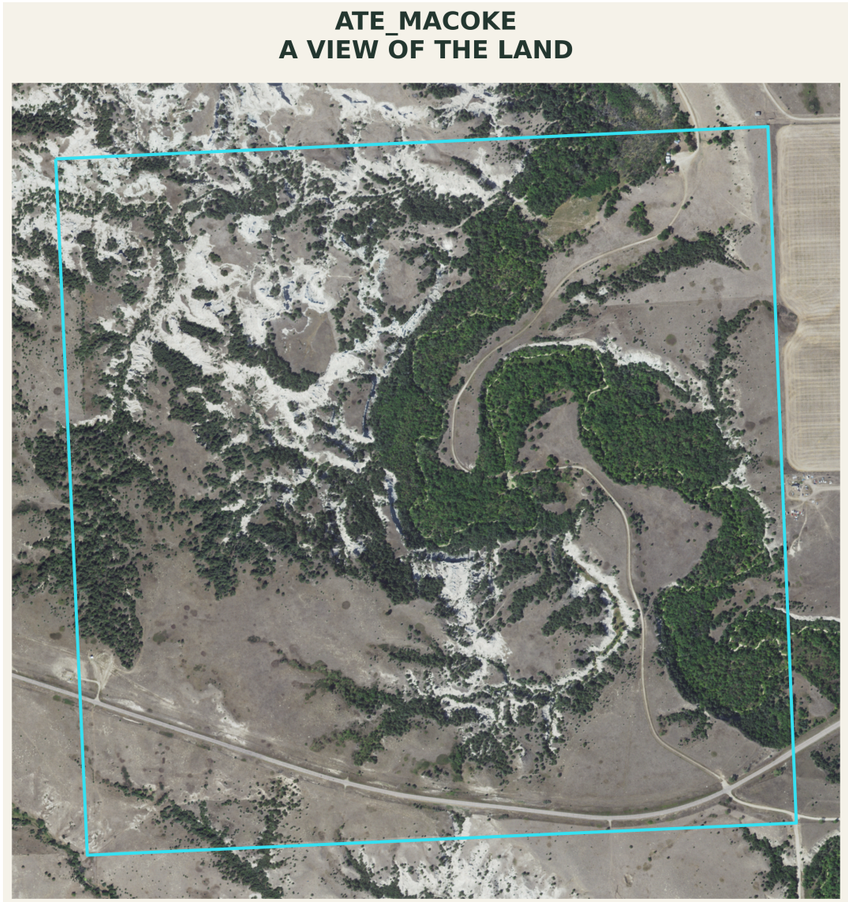
This overview brings together several views of the land. Each view helps us consider a different part of the assessment.

WHAT DO YOU NOTICE?

Which places would you like to understand better?

MAP 02

A View of the Land



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

READING THE MAP

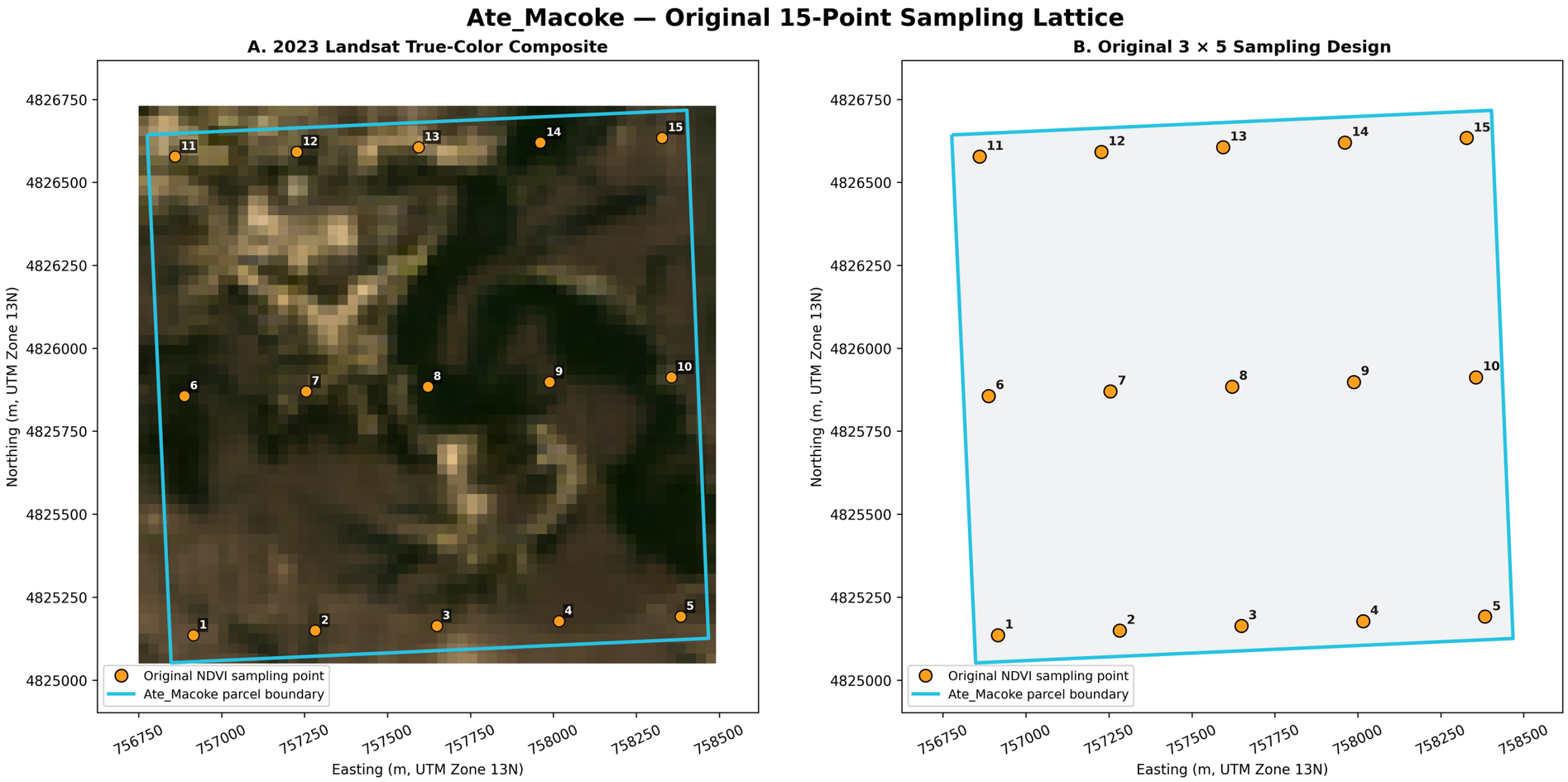
This aerial view helps us recognize the parcel, its boundary, and features of the surrounding landscape.

WHAT DO YOU NOTICE?

What places or features do you recognize?

MAP 03

Where We Looked



Original 15-point lattice (2015–2023) | Ate_Macoke parcel: 258.82 ha | Landsat 2023 RGB: 30 m

READING THE MAP

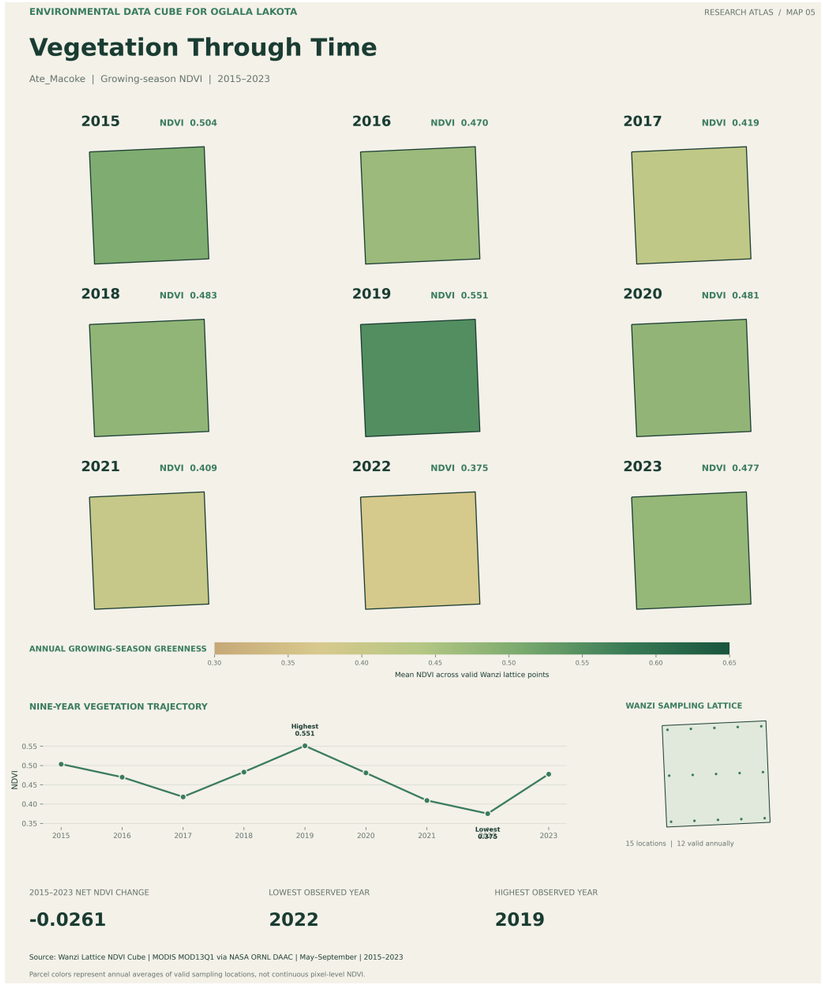
These sampling locations show where vegetation observations were organized for the assessment.

WHAT DO YOU NOTICE?

Are there other places that should be examined?

MAP 04

Vegetation Through Time



READING THE MAP

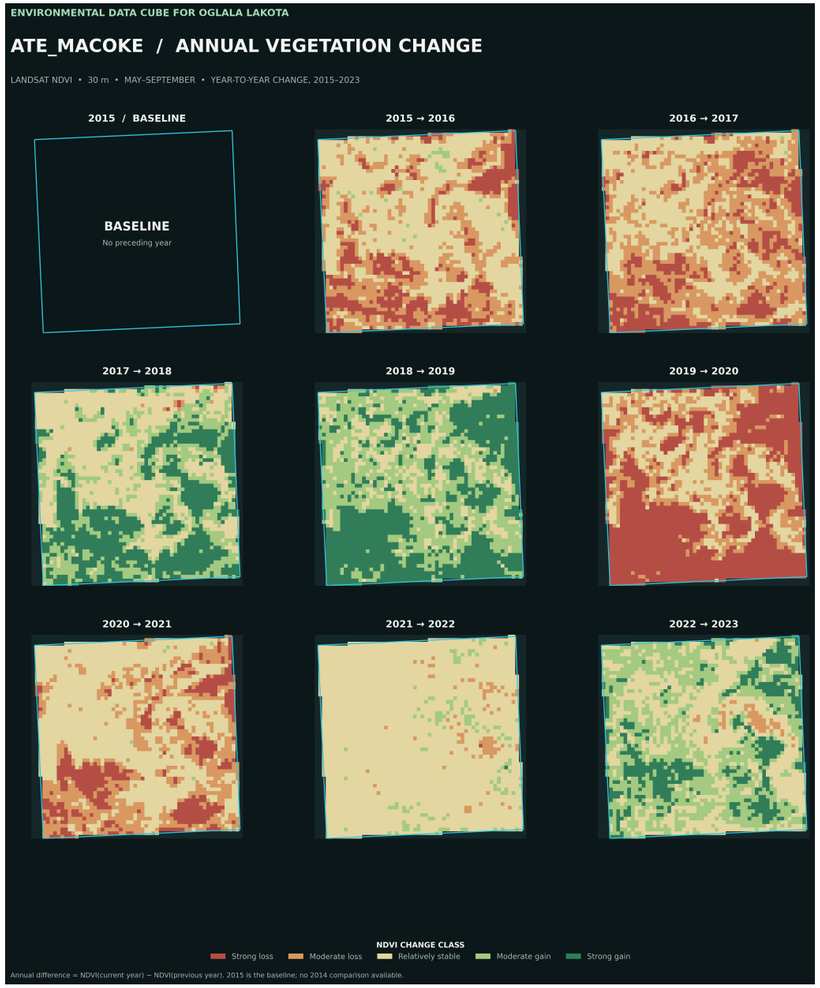
This map shows how vegetation greenness varied across the assessment years. Weather and seasonal growth can influence what the satellite observes.

WHAT DO YOU NOTICE?

Which years stand out? Do they match what you remember?

MAP 05

How Vegetation Changed



READING THE MAP

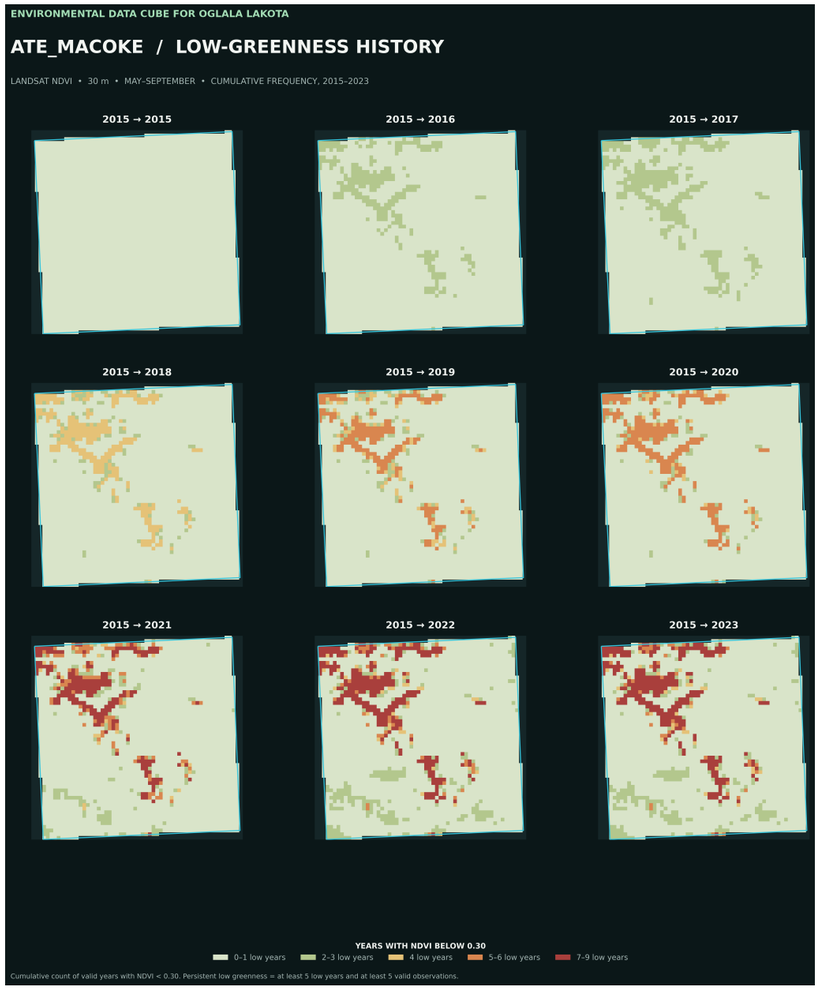
Comparing vegetation observations over time helps us identify places where greenness may have increased or decreased.

WHAT DO YOU NOTICE?

Do these patterns match changes you have seen?

MAP 06

Places With Lower Greenness



READING THE MAP

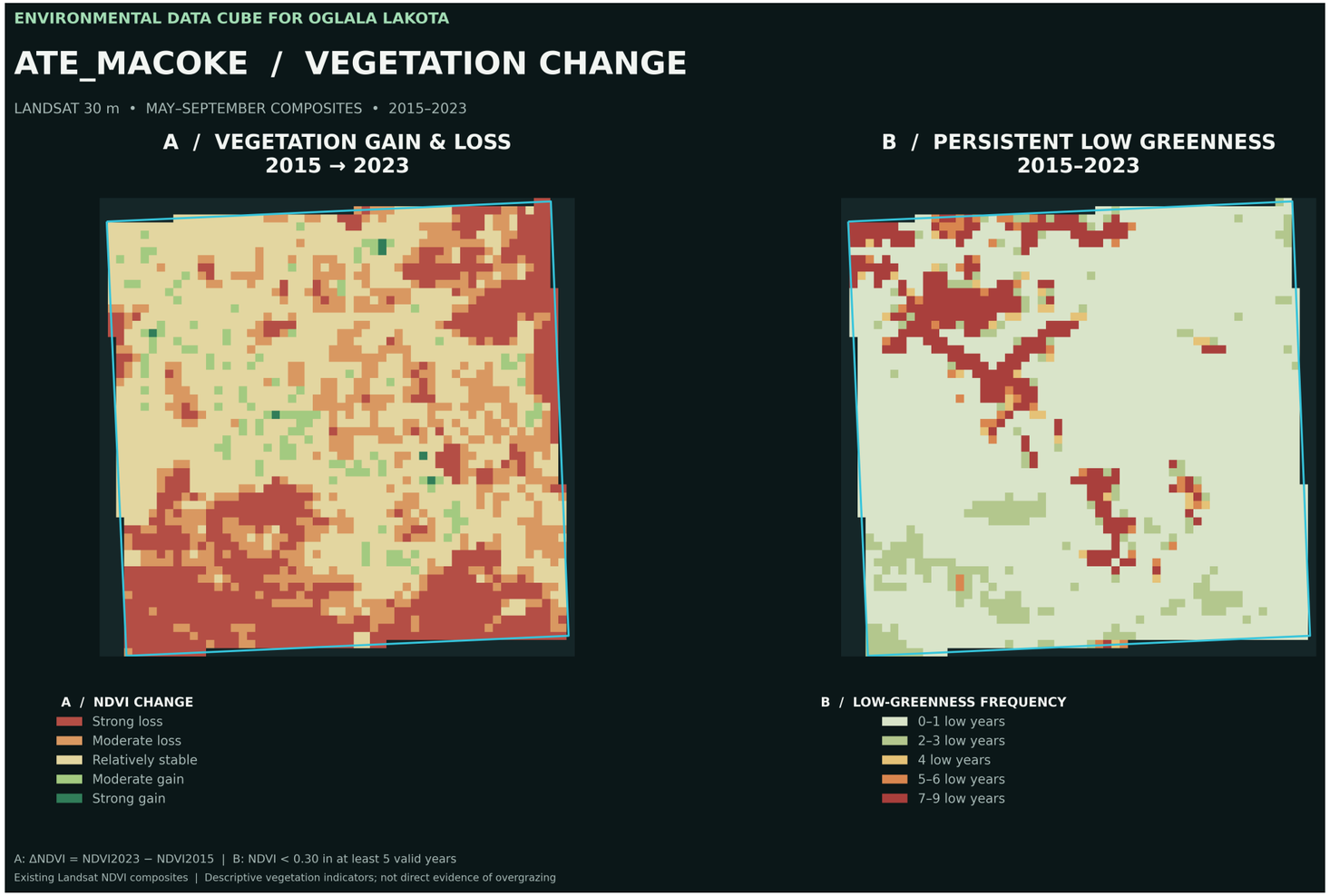
This map highlights recurring patterns of lower vegetation greenness. Lower greenness alone does not establish overgrazing or land degradation.

WHAT DO YOU NOTICE?

What do you know about these places?

MAP 07

Change and Persistence



READING THE MAP

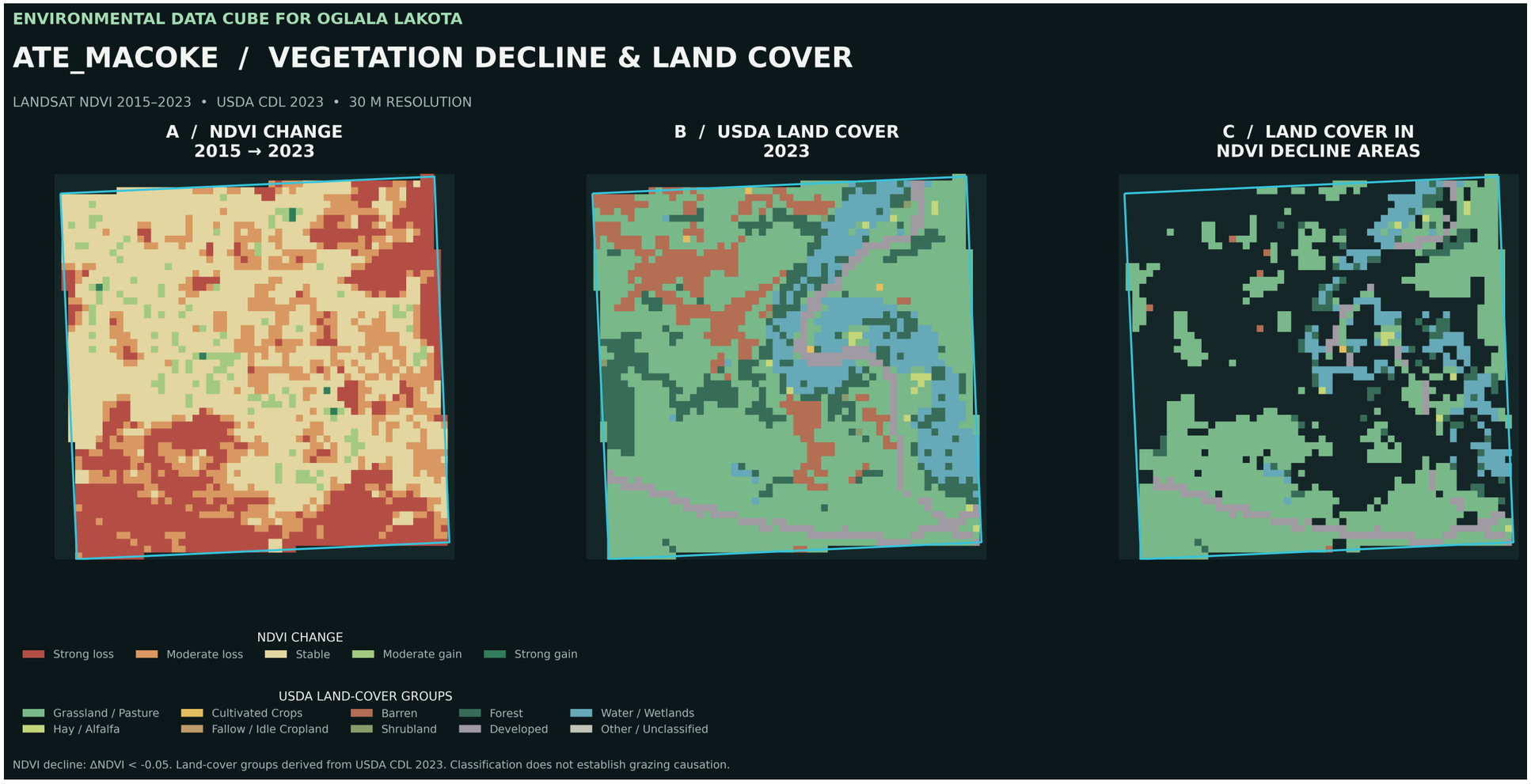
This map helps compare places where vegetation patterns changed with places showing more persistent conditions.

WHAT DO YOU NOTICE?

Which areas would benefit from closer observation?

MAP 08

Where Should We Look Closer?



READING THE MAP

Vegetation and land-cover observations can help guide questions about the landscape, but field observations remain important.

WHAT DO YOU NOTICE?

Where would you begin a visit to the land?

Understanding the Results

Vegetation greenness is an indicator observed by satellites. It does not directly measure soil health or grazing pressure.

Lower greenness does not by itself establish overgrazing or land degradation.

Aerial photographs and vegetation observations may have different collection dates.

Field observations and knowledge of the land are needed to interpret the patterns.

Confirm map details and sharing permissions before distributing this community review draft.

Observations From the Land

Places I recognize

Changes I have noticed

Questions about the maps

Places to visit together